Chemical Vapor Deposition (CVD) System

Equipment Technical Documentation (Sample)

---

1. Equipment Overview

The Chemical Vapor Deposition (CVD) system is used for thin film deposition on semiconductor wafers. The system operates by introducing precursor gases into a reaction chamber, where controlled temperature and pressure enable film formation.

Tool Type: CVD
Model: CVD-X100
Communication Protocol: SECS/GEM (HSMS)

---

2. Process Parameters (Variables)

2.1 Status Variables (SV)

VID	Name	Type	Unit	Description
V1001	ChamberTemperature	Float	°C	Current chamber temperature
V1002	ChamberPressure	Float	Torr	Current chamber pressure
V1003	GasFlowRate	Float	sccm	Gas flow rate
V1004	RFPower	Float	W	RF power applied
V1005	ProcessStatus	String	-	Current process state

2.2 Data Variables (DV)

VID	Name	Type	Unit	Description
V2001	RecipeID	String	-	Active process recipe
V2002	WaferCount	Integer	-	Number of wafers processed
V2003	DepositionTime	Float	sec	Total deposition duration

---

3. Equipment Events (CEID)

CEID	Event Name	Description	Linked VIDs
E101	ProcessStart	Triggered when process begins	V1001, V1002, V2001
E102	ProcessEnd	Triggered when process completes	V2002, V2003
E103	WaferLoaded	Wafer loaded into chamber	V2001
E104	WaferUnloaded	Wafer removed from chamber	V2002
E105	AlarmTriggered	Alarm condition detected	V1001, V1002

---

4. Alarm Definitions

Alarm ID	Name	Severity	Description
A201	HighTemperature	Critical	Temperature exceeds safe limit
A202	LowPressure	Warning	Pressure below threshold
A203	GasFlowError	Critical	Gas flow instability detected

---

5. Remote Commands (RCMD)

Command	Description	Parameters
START	Start process	RecipeID
STOP	Stop process	None
PAUSE	Pause process	None
RESUME	Resume process	None

---

6. Equipment States

State ID	State Name	Description
S1	Idle	System ready, no process
S2	Processing	Active deposition process
S3	Completed	Process completed
S4	Error	Fault or alarm condition

---

7. State Transitions

- Idle → Processing (Triggered by START command)
- Processing → Completed (Triggered by ProcessEnd event)
- Processing → Error (Triggered by AlarmTriggered event)
- Error → Idle (After reset)

---

8. Communication Details

- Protocol: SECS/GEM
- Transport: HSMS (TCP/IP)
- Port: 5000

Example Messages:

- S1F1 → Establish communication
- S6F11 → Event report (ProcessStart, ProcessEnd)
- S2F41 → Remote command (START, STOP)

---

9. Notes

- All variables are updated in real-time during processing
- Alarm conditions must be acknowledged before reset
- Recipe parameters are configurable via MES